

UNIT - 1

COMPUTER NETWORK

Computer Network. 2023

A computer network is a system in which two or more computers/devices are connected with each other through communication links to share data, resources, and services.

Main Points.

- Collection of Interconnected devices
- Uses communication channels (wired/wireless)
- Follows Protocols (rules)
- Enable data exchange.

Example → Internet / College Wi-Fi network.

Goals of Computer Network.

The main objectives (goals) of a computer network are :-

1. Resource Sharing

- Share Hardware - printer, scanner
- Share software - applications
- Share data - files, documents,

Example - One printer used by many computers.

2. Communication - Exchange information easily

- Email, chat, video calls.

→ Makes communication fast and global.

3. Reliability (High Availability)

- Data can be accessed from multiple systems.

- Backup copies available.

→ If one system fails, another can work.

4. Cost Reduction

- Sharing resources reduces cost
- No need to buy separate devices for each other.

5. Scalability (Expansion)

- Easy to add more computers.
- Network can grow without major changes.

Components of Computer Network.

A computer network consists of the following basic components:-

① Sender (source).

Device that sends data

Example → Computer, mobile.

③ Message (Data)

- Information to be transmitted
- Text, image, video.

④ Transmission Medium

Path through which data travels.

Types: -

- Wired - cable, fiber
- Wireless - wifi, radio waves.

⑤ Protocols.

Set of rules for communication

Functions -

- Data format
- Error Handling
- Transmission control.

Examples - TCP, HTTP

⑥ Network Devices.

Router - Connect different networks

Switch - Connect devices in a LAN

Hub - Simple devices (broadcasts data).

Architecture of Computer Network.

Network architecture refers to the design and structure of a computer network, including.

- How devices are connected
- How data flows
- How communication happens

Types of Network Architecture.

There are mainly two types :-

① Client - server Architecture.

In this, a client server provides services / resources, and multiple clients request those services.

- **server** - Powerful system (stores data, controls networks)
- **Clients** User devices (request services)

Working.

- 1 Client sends request.
- 2 Server processes request.
- 3 Server sends response.

Advantages

- Centralized control
- Better security
- Easy management
- Data backup possible

Disadvantages

- Server failure $\rightarrow$ whole network stops
- Expensive to maintain
- High traffic on server

Example

Banking system, Email server, websites.

② Peer-to-Peer (P2P) Architecture

In this architecture, all computers are equal and can act as both client and server.
Structure.

- No central server
- Each node shares resources directly.

Working.

One computer can send and receive data directly from another.

Advantages

- Low cost
- Easy to set up
- No need of dedicated server

Disadvantages

- Less security
- ~~Easy~~ ^{Hard} to set up
- No need of dedicated server.

Example

- File sharing system
- Small office networks.

Classification And Types of Computer Network.

- ① LAN (Local Area Network)
- ② MAN (Metropolitan Area Network)
- ③ WAN (Wide Area Network)

These are classified based on the geographical area covered.

① LAN (Local Area Network).

A Local Area Network (LAN) is a computer network that connects devices within a small geographical area such as a room, office, building, or campus to share data and resources.

Characteristics

- It covers small area
- High data transfer speed
- It's cost is low.
- Usually privately owned
- Easy to install and maintain.

Components of LAN

A LAN consists of:-

- ① → Computers (Nodes)
Devices like PCs, laptops, mobiles.
- ② → Network Devices
 - Switch
 - Router
 - Hub.
- ③ → Transmission Media.
 - Wired → Ethernet cable
 - Wireless → Wi-Fi.
- ④ → Network Interface Card (NIC)
 - Allows computer to connect to network.

Advantages of LAN

- Faster data transfer compared to WAN.
- Share printer, files, Internet.
- Installation and maintenance are cheap
- Quick file transfer and messaging.

Disadvantages of LAN

- Cannot cover large distances
- Data can be accessed if not secured
- If main device fails, network may stop.

Applications of LAN

- School and colleges.
- Offices
- Computer labs
- Home network.

Example → A college lab where 20 computers are connected and share one printer is a LAN.

② MAN (Metropolitan Area Network)

A metropolitan Area Network (MAN) is a network that connects multiple computers or LANs over a large area like a city or metropolitan region.

Characteristics.

- Covers city-level area.
- Larger than LAN, smaller than WAN
- Medium data speed.
- Can be public or private.
- Connects multiple LANs together.

Advantages of MAN

- It covers entire city
- Connects multiple LANs
- Fast communication within city

Disadvantages of MAN

- More expensive than LAN
- It is difficult to maintain
- Data may be at risk.

Application of MAN

- City-wide internet services
- Cable TV networks.
- Government communication systems
- Smart city network.

Examples - Internet service providers connecting different areas of a city is a MAN.

③ WAN (Wide Area Network)

A Wide Area Network is a computer network that connects devices over a very large geographical area, such as countries or continents.

Characteristics

- Connect very large area.
- Connect multiple LANs and MANs.
- Uses public communication systems (satellite,

telephones lines . internet) .

- low speed compared to LAN
- High cost.

Advantages

- connects people worldwide
- covers countries and continents

Disadvantages.

- Installation and maintenance expensive
- Slower than LAN
- More chances of data hacking.

Applications of WAN

- Internet
- Banking Network
- Airline reserve system.
- Government network.

Example- The internet is the largest WAN connecting the world wide.

LAYERED ARCHITECTURE

Layer architecture means breaking the complex task of communication into small layers, where each layer performs a specific function.

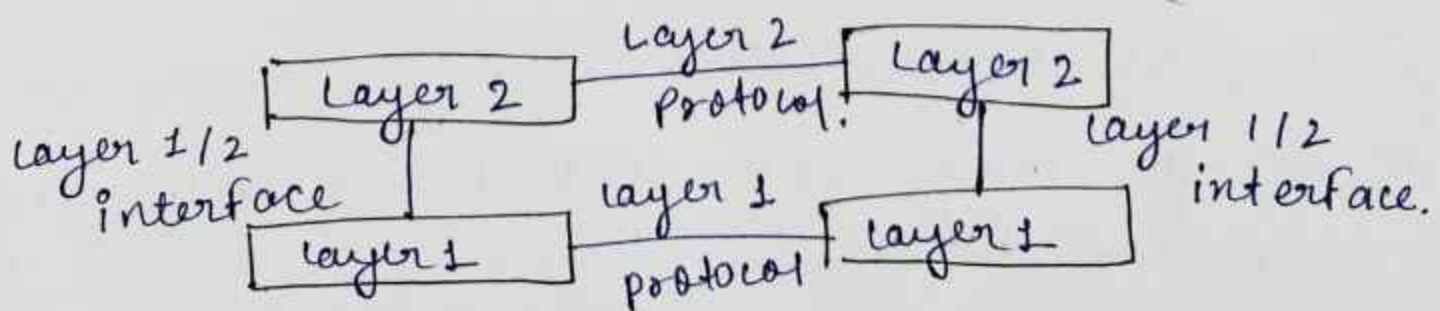
Example → Instead of doing everything at once, the network is divided into layers like a team doing different tasks.

PROTOCOL HIERARCHY

A protocol hierarchy is the arrangement of different layers, where..

Each layer uses its own protocol.

Each layer communicates with the same layer on another system.



Layer architecture of computer network design.

→ Layer architecture have several advantages.

Some of them are -

- 1 Modularity and clear interface
- 2 Provides flexibility to modify network services.
- 3 Ensure independence of layers.
- 4 Management of network architecture is easy.

Design Issues.

→ When designing layers, some important problem issues must be solved.

- 1 Every device must have a unique address
Eg → IP address.
- 2 Errors may occurs during transmission.
- 3 Sender may send data faster than receiver can handle.

- Large data is divided into smaller parts (packets).

Interfaces and Services:-

Interface

→ An Interface is the point of interaction between two layers.

It defines how the upper layer can use the services of the lower layer.

Example → Transport layer gives data transfer service.

• Interface tells how application layer sends / receive data.

Service

A service is the function or facility that a layer provides to the layer above it.

lower layer → provides help to upper layer.

Types of Services. 2025

1. Connection - Oriented Service.
2. Connection-less service

① Connection-Oriented Service. 2024

A Connection-oriented service is a type of communication where a connection is established before sending data. They are:

1. Connection is established
2. Information is sent
3. Connection is released.

like making a phone call.

- First you connect
- Then you talk
- Then you disconnect.

Working steps.

- 1 sender and receiver create a connection.
- 2 Data is sent in sequence.
- 3 ~~error~~ connection is closed after communication.

Features →

- 1 Reliable communication
- 2 Data delivered in correct order
- 3 Error checking and correction.
- 4 Flow control is present.

② Connectionless service.

A connectionless service is a type of communication where no connection is established before sending data.

- Like sending a message.
- No setup.
- Just send data directly.

Working

- 1 Data is sent as independent packets.
- 2 Each packet may take a different path
- 3 No guarantee of delivery.

Features

- 1 Fast communication
- 2 No connection setup
- 3 No guarantee of order.
- 4 No error recovery.

SERVICE PRIMITIVES. 2023

Service primitives are basic operations or commands that a layer provides to the upper layer to access its services.

They are the actions used to use a service.

Types of Service Primitives.

There are five main primitives.

- ① LISTEN - Wait for an incoming connection, used by server.

Example → server wait for client request.

- ② CONNECT - Establish a connection with ~~out~~ another device. used by client.

Example → client requests connection to server.

- ③ SEND → send data to the receiver.
like sending message / File.

- ④ RECEIVE → Receive data from sender.
Example → Reading incoming message.

- ⑤ DISCONNECT → Terminate the connection
Example → Ending communication.

→ Need of Service Primitives.

- To establish connection
- To send and receive data
- To terminate connection.
- To control communication between layers.

★ ISO OSI Model (Open Systems Interconnection).
2024, 2025
2023

The OSI Model (Open System Interconnection Model) is a 7-layer conceptual framework that describes how data moves from one device to another over a network.

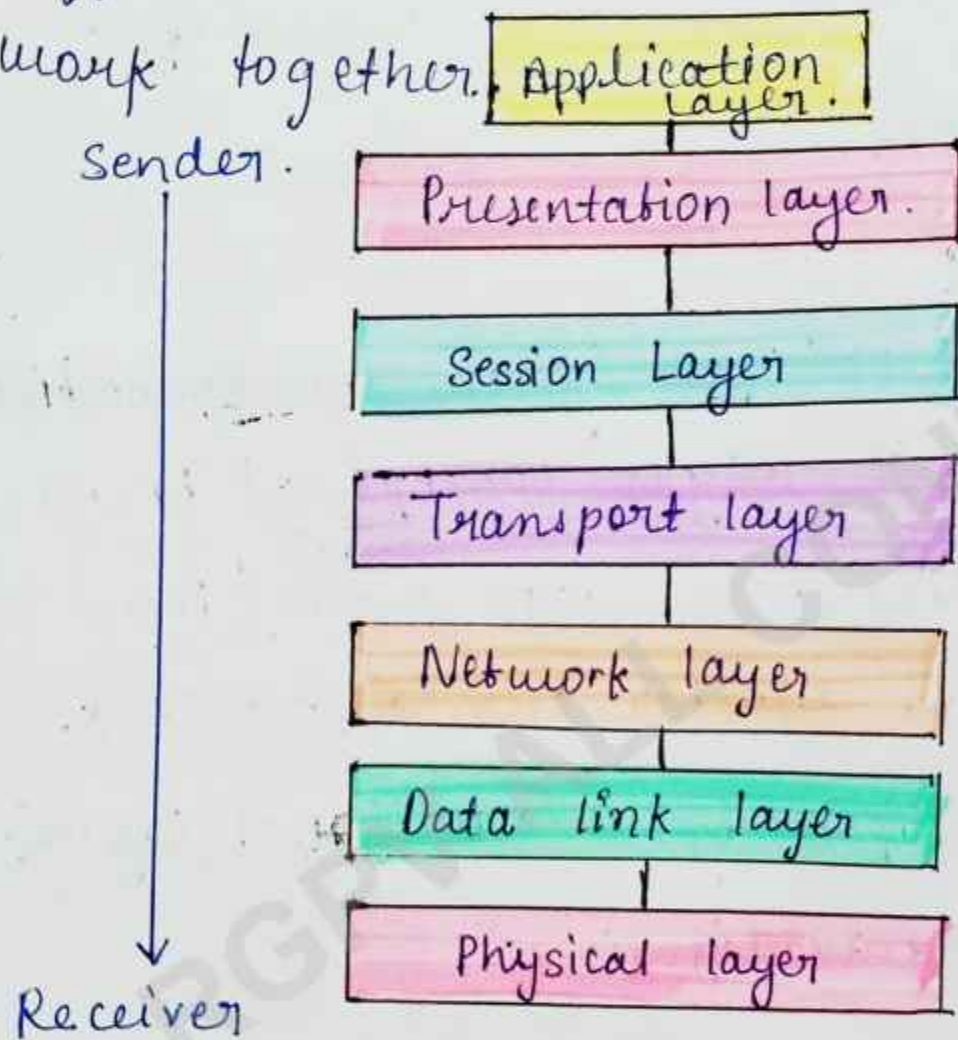
It was Developed by International Organization for Standardization.

The main Goal of OSI Model is

- Standardize Communication
- And Different systems can communicate easily.

The each OSI layer has specific functions and responsibilities.

This layered approach makes it easier for different devices and technologies to work together.



OSI Model.

① Application Layer (Layer 7)

It is the top most layer of OSI model. This layer provides services directly to the user/application.

- This provides Interface between user and network.
- provides services like, web browsing, email, file transfer.
- Use protocols → HTTP/HTTPS, FTP, SMTP.

② Presentation Layer (Layer 6)

- This layer Acts as a translator between network.
- Handles data translation, encryption, and compression to ensures that the application layer can understand the received data.

③ Session Layer (Layer 5)

- It manages the connection / session between two devices.
- Handling the opening, maintaining, and closing of communication.

④ Transport layer (Layer 4)

- Responsible for end to end communication and error recovery.
- It breaks data into segments and ensures they arrives correctly using protocols like TCP & UDP.

⑤ Network layer (Layer 3)

- It handles best data routing (path) between different networks.
- It packages segments into packets and uses logical addressing (IP addresses) to find the best path.

⑥ Data Link Layer (Layer 2)

- Manages data transfer between two devices on the same network.
- It uses physical addresses (MAC address).

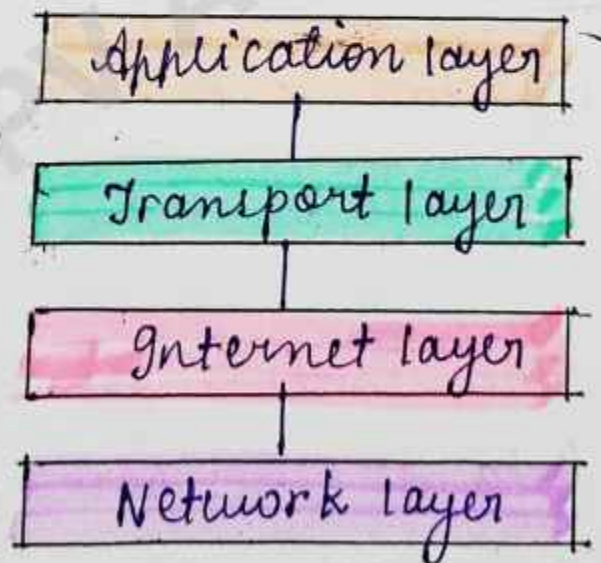
⑦ Physical layer (Layer 1)

- It is the lowest layer of the OSI model.
- Handles the actual transmission of data.
- It transmits raw bits (0s and 1s) over physical media like cables, fiber optics, or radio waves (Wi-Fi).

TCP / IP Model 2024

(Transmission Control Protocol / Internet Protocol)

- TCP/IP model is a 4-layered conceptual framework - also known as Internet Protocol Suite or DoD Model that defines how data is packetized, addressed, transmitted, routed and received over network.
- It is the actual networking model used on the internet today.
Developed by United States Department of Defence.



TCP / IP Model.

★ Application layer.

- This is the top most layer and closest to the user.
- It combines three OSI layers.
- ★ Application, Presentation and Session layer.
- Provides network services to applications. like web browsing, email, file transfer.
- Provide data representation (text, video, images).
- Protects sensitive data (Encryption & Security)

★ Transport Layer.

It manage end to end communication and data delivery.

Protocols.

TCP (Transmission control protocol)

- Reliable (data reaches correctly).
- Connection oriented

UDP (User Datagram Protocol)

- Faster but unreliable
- Connectionless.

★ Internet layer

- Handles logical addressing and routing (path) of data.
- It finds the best path to send data from source to destination.

Protocols.

- IP (Internet Protocol).
- ICMP (for error messages).
- ARP (address resolution)

★ Network Layer

- This is the lowest layer and deals with the physical transmission of data.
- It includes framing, hardware (MAC) addressing, and Ethernet / wifi communication.

Merits / Advantages of TCP/IP Model.

1. It is not owned by any company so widely accepted and used all over world.
2. Can easily handle millions of devices.
3. Works with Ethernet, Wi-Fi, Fiber optic, and flexible with different hardware.
4. Ensures data reaches correctly and in order.
5. Uses IP to find best path for data transmission.

Demerits / Disadvantages of TCP/IP Model.

1. Configuration and management can be difficult for beginners.
2. Unlike OSI, boundaries between layers are not well defined.
3. Basic TCP/IP does not provide strong security so it needs additional protocols (like SSL/TLS).

4. Designed mainly for Internet communication, so less suitable for other types of network.

Difference between OSI and TCP model.

OSI

① OSI (Open System Inter-connection) model is a theoretical framework developed by ISO.

② OSI model has seven layers

③ It is protocol-independent. does not depend on any protocol.

④ OSI used for learning concepts and troubleshooting. not used in real implementation

⑤ More structured and strict

TCP

① The TCP/IP model is a practical model developed by DARPA.

② TCP/IP model has four layers

③ It is protocol dependent. Built around TCP, IP, UDP, HTTP.

④ TCP/IP used in internet, networking system and real-world implementation.

⑤ More Flexible.

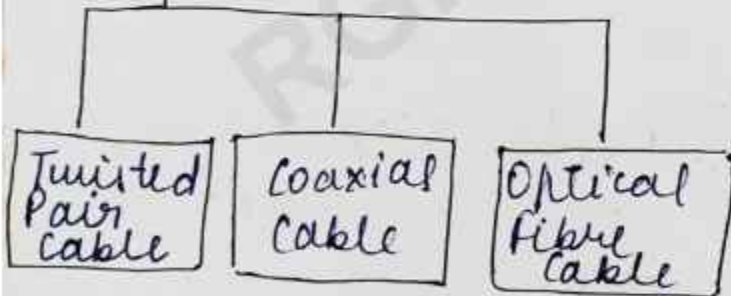
MEDIA | TRANSMISSION MEDIA

Transmission media is the communication path through which data travels from sender to receiver.

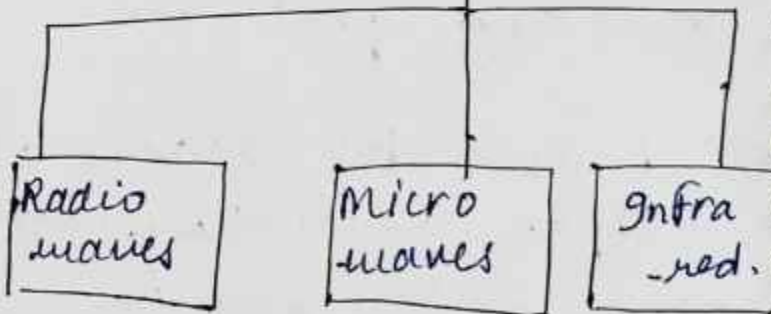
media = medium / road
used to send
data.

Types of Transmission Media

Guided media



Unguided media



Transmission media is mainly divided into 2 types.

- ① Guided media - Data travels through physical path. (cables).

(a) Twisted pair - Two insulated copper wires twisted together.

- used in telephone lines and LAN.
- It is cheap and easy to install.
- But more noise and interference.

(b) Coaxial cable

- It has central conductor and insulation shielding.
- used in TV cables network.
- Provide better protection from noise.
- Higher bandwidth than twisted pair.
- But expensive than Twisted pair.

(c) Optical Fiber cable.

- Uses light signal instead of electrical signals.
- Made of glass / plastic fibers.
- It has very high speed.
- No electromagnetic interference & provide long distance communication.
- But it is expensive and Difficult installation

② Unguided Media. (Wireless / Unbounded).

Data travels through air using electromagnetic waves.

(a) Radio waves.

- Used in radio, TV, mobile communication.
- long distance covered and penetrate walls.
- low security.

(b) Microwaves.

- Used in satellite communication, mobile towers.
- It has high data rate

(c) Infrared.

- Used in remote controls, short range communication
- Secure (can not pass through walls)
- Very short range.

BANDWIDTH

Bandwidth is the maximum capacity of a communication channel to transmit data.

Bandwidth $\rightarrow$ how much data can be sent per unit time.

Types of Bandwidth.

① Analog Bandwidth $\rightarrow$ It is measured in Hertz (Hz).

Refers to the range of frequencies a signal can occupy.

Example.

$\rightarrow$ If a channel support 1000 Hz to 5000 Hz
Bandwidth = 4000 Hz.

② Digital Bandwidth.

Measured in bits per second (bps)

Refers to data transmission capacity.

Example:-

$\rightarrow$ 100 mbps connection means
100 million bits per second.

DATA RATE

Data rate is the actual amount of data transmitted per unit time over a network.

Data rate - speed at which data is sent.
Or.

Data rate is the number of bits transmitted per second over a communication channel.

Units of Data rate

- Bits per second (bps).
- Kbps = 10^3 bps
- Mbps = 10^6 bps
- Gbps = 10^9 bps

Types of Data Rate.

① Bit Rate - Number of bits transmitted per second.

Example - 100 Mbps $\rightarrow$ 100 million bits per second.

② Band Rate (Symbol Rate)

- Number of signal units (symbols) transmitted per second.

MODULATION

Modulation is the process of converting digital data (0s and 1s) into signals that can be transmitted over a communication medium like cables or air (wireless).

Why modulation is Needed.

Computers generate digital signals, but many transmission media (like radio waves) work together with analog signals.

So, modulation helps.

- Send data over long distances
- Reduce noise and interference.
- Efficiently use bandwidth.

Basic Idea.

In modulation

- A carrier signal (high-frequency wave) is used.
- The data signal modifies this carrier.